

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397843
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Haut; Robert E. et al.

Stroller and basket thereof

Abstract

A stroller includes a frame and a basket disposed on the frame. The basket includes a tube mount, a first tube member, a second tube member and a basket covering. The tube mount is disposed on the frame. The first tube member is connected to the tube mount. The second tube member is connected to the tube mount and rotatably switchable relative to the frame between a folded position and an unfolded position. The basket covering is disposed on the first tube member and the second tube member. A storage space is enclosed by the basket covering. A volume of the storage space is adjustable by rotation of the second tube member relative to the frame.

Inventors: Haut; Robert E. (Merion Station, PA), Eggert-Crowe; Colin F. (Philadelphia, PA)

Applicant: Wonderland Switzerland AG (Steinhausen, CH)

Family ID: 1000008778205

Assignee: Wonderland Switzerland AG (Steinhausen, CH)

Appl. No.: 18/220834

Filed: July 12, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230347962 A1	Nov. 02, 2023

Related U.S. Application Data

continuation parent-doc US 17156659 20210125 US 11738792 child-doc US 18220834
continuation parent-doc US 16563955 20190909 US 10940880 20210309 child-doc US 17156659
us-provisional-application US 62729115 20180910

Publication Classification

Int. Cl.: B62B9/26 (20060101); B62B7/06 (20060101); B62B9/20 (20060101)

U.S. Cl.:

CPC B62B9/26 (20130101); B62B7/06 (20130101); B62B9/20 (20130101);

Field of Classification Search

CPC: B62B (9/26); B62B (7/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7458599	12/2007	Kassai	N/A	N/A
7500692	12/2008	Espenshade	N/A	N/A
9090277	12/2014	Chen	N/A	N/A
9402774	12/2015	Martin	N/A	N/A
10940880	12/2020	Haut	N/A	B62B 9/20
11738792	12/2022	Haut	280/47.38	B62B 9/20
2004/0070178	12/2003	Yoshie	N/A	N/A
2004/0222608	12/2003	Espenshade	N/A	N/A
2005/0161912	12/2004	Cheng	N/A	N/A
2005/0225056	12/2004	Dotsey	N/A	N/A
2006/0214397	12/2005	Dotsey	N/A	N/A
2007/0216138	12/2006	Yang	N/A	N/A
2008/0258437	12/2007	Ryan	N/A	N/A
2011/0175309	12/2010	Newhard	N/A	N/A
2014/0183843	12/2013	Iftinca	N/A	N/A
2018/0022371	12/2017	Jakubowski	N/A	N/A
2021/0022314	12/2020	Xiang	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2808687	12/2005	CN	N/A
101691123	12/2009	CN	N/A
102556133	12/2011	CN	N/A
106949353	12/2016	CN	N/A
3 159 239	12/2016	EP	N/A
2005-162047	12/2004	JP	N/A
2009-173075	12/2008	JP	N/A
2011/087506	12/2010	WO	N/A

Primary Examiner: Swenson; Brian L

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/156,659, filed on Jan. 25, 2021, which is a continuation application of U.S. application Ser. No. 16/563,955, filed on Sep. 9, 2019, which claims the benefit of U.S. Provisional Application No. 62/729,115, filed on Sep. 10, 2018. The contents of these applications are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a stroller and a basket thereof, and more particularly, to a stroller and a basket thereof with expandable storage space.

2. Description of the Prior Art

(2) Most strollers are used from birth to about three years old to transport babies and young children. Caring for an infant or toddler requires a lot of equipment and supplies, such as diapers, diaper bags, snacks, spare clothing, toys, etc. A conventional stroller usually has a fixed fabric basket located under a seat for storing the equipment and the supplies. However, most often there is not enough storage space. Sometimes a parent or a caregiver will resort to piling coats on a canopy or hanging a bag off back from a handle, which makes the stroller prone to tipping over.

SUMMARY OF THE INVENTION

(3) Therefore, it is an objective of the present invention to provide a stroller and a basket thereof with expandable storage space for solving the aforementioned problem.

(4) In order to achieve the aforementioned objective, the present invention discloses a stroller including a frame and a basket disposed on the frame. The basket includes a tube mount, a first tube member, a second tube member and a basket covering. The tube mount is disposed on the frame. The first tube member is connected to the tube mount. The second tube member is connected to the tube mount and rotatably switchable relative to the frame between a folded position and an unfolded position. The basket covering is disposed on the first tube member and the second tube member. A storage space is enclosed by the basket covering, and a volume of the storage space is adjustable by rotation of the second tube member relative to the frame.

(5) In order to achieve the aforementioned objective, the present invention further discloses a basket adapted for a frame of a stroller. The basket includes a tube mount, a first tube member, a second tube member and a basket covering. The tube mount is disposed on the frame. The first tube member is connected to the tube mount. The second tube member is connected to the tube mount and rotatably switchable relative to the frame between a folded position and an unfolded position. The basket covering is disposed on the first tube member and the second tube member. A storage space is enclosed by the basket covering, and a volume of the storage space is adjustable by rotation of the second tube member relative to the frame.

(6) In summary, in the present invention, when the second tube member is rotated from the folded position to the unfolded position, the volume of the storage space enclosed by the basket covering is increased. When the second tube member is rotated from the unfolded position to the folded position, the volume of the storage space enclosed by the basket covering is decreased. Therefore, it allows a user or a caregiver to easily adjust the volume of the storage space according to practical demands by rotating the second tube member relative to the frame, which brings convenience in use.

(7) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of a stroller according to an embodiment of the present invention.
- (2) FIG. 2 and FIG. 3 are lateral view diagrams of the stroller in different states according to the embodiment of the present invention.
- (3) FIG. 4 and FIG. 5 are partial diagrams of the stroller at different views according to the embodiment of the present invention.
- (4) FIG. 6 is a partial diagram of a basket according to the embodiment of the present invention.
- (5) FIG. 7 is a partial exploded diagram of the basket according to the embodiment of the present invention.
- (6) FIG. 8 is a partial internal structural diagram of the basket according to the embodiment of the present invention.
- (7) FIG. 9 and FIG. 10 are partial internal structural diagrams of the basket in different states according to the embodiment of the present invention.

DETAILED DESCRIPTION

(8) In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration specific embodiments in which the invention may be practiced. In this regard, directional terminology, such as “top,” “bottom,” “front,” “back,” etc., is used with reference to the orientation of the Figure(s) being described. The components of the present invention can be positioned in a number of different orientations. As such, the directional terminology is used for purposes of illustration and is in no way limiting. Accordingly, the drawings and descriptions will be regarded as illustrative in nature and not as restrictive.

(9) Please refer to FIG. 1 to FIG. 5. FIG. 1 is a schematic diagram of a stroller 1 according to an embodiment of the present invention. FIG. 2 and FIG. 3 are lateral view diagrams of the stroller 1 in different states according to the embodiment of the present invention. FIG. 4 and FIG. 5 are partial diagrams of the stroller 1 at different views according to the embodiment of the present invention. As shown in FIG. 1 to FIG. 5, the stroller 1 includes a frame 11 and a basket 12 disposed on the frame 11. The basket 12 includes two tube mounts 121, a first tube member 122, a second tube member 123 and a basket covering 124. The two tube mounts 121 are disposed on the frame 11 and located at two lateral sides of the frame 11. The first tube member 122 is connected to the two tube mounts 121. The second tube member 123 is rotatably connected to the two tube mounts 121 and switchable relative to the frame 11 between an unfolded position as shown in FIG. 2 and a folded position as shown in FIG. 3. The basket covering 124 is disposed on the first tube member 122 and the second tube member 123. A storage space is enclosed by the basket covering 124, and a volume of the storage space is adjustable by rotation of the second tube member 123 relative to the frame 11. When the second tube member 123 is rotated from the folded position as shown in FIG. 3 to the unfolded position as shown in FIG. 2, the basket covering 124 is gradually expanded, so that the volume of the storage space enclosed by the basket covering 124 is increased. When the second tube member 123 is rotated from the unfolded position as shown in FIG. 2 to the folded position as shown in FIG. 3, the basket covering 124 is gradually collapsed, so that the volume of the storage space enclosed by the basket covering 124 is decreased. However, the number of the tube mount is not limited to this embodiment. For example, in another embodiment, there can be only one tube mount located at one side of the frame.

(10) Please refer to FIG. 1 to FIG. 10. FIG. 6 is a partial diagram of the basket 12 according to the embodiment of the present invention. FIG. 7 is a partial exploded diagram of the basket 12 according to the embodiment of the present invention. FIG. 8 is a partial internal structural diagram of the basket 12 according to the embodiment of the present invention. FIG. 9 and FIG. 10 are

partial internal structural diagrams of the basket **12** in different states according to the embodiment of the present invention. As shown in FIG. **1** to FIG. **10**, the two tube mounts **121** located at the two lateral sides of the frame **11** can have the same structure. The tube mount **121** at one lateral side of the frame **11** is introduced as follows for simplicity. The tube mount **121** includes a first mount member **1211** and a second mount member **1212**. The first mount member **1211** is disposed on the frame **11** and connected to the first tube member **122**. The second mount member **1212** is disposed on the second tube member **123** and rotatably combined with the first mount member **1211**. The second tube member **123** is rotatable relative to the frame **11** by rotation of the second mount member **1212** relative to the first mount member **1211**.

(11) Furthermore, the first mount member **1211** includes a housing **1211A**. The second mount member **1212** includes a hub **1212A** rotatably combined with the housing **1211A**. A plurality of first ribs **1211B** is formed on at least one portion of an inner wall of the housing **1211A**. A plurality of second ribs **1212B** is formed on at least one portion of an outer wall of the hub **1212A**. The first mount member **1211** and the second mount member **1212** engage with each other by the plurality of first ribs **1211B** and the plurality of second ribs **1212B** for positioning the second tube member **123** at a desired position between the folded position and the unfolded position. In other words, the second tube member **123** can be positioned at any position between the unfolded position and the folded position by positioning the second mount member **1212** relative to the first mount member **1211** with engagement of the plurality of first ribs **1211B** and the plurality of the second ribs **1212B**, which achieves a purpose of multistage adjustment of the volume of the storage space.

(12) Preferably, in this embodiment, the plurality of first ribs **1211B** can be formed around at least one portion of the inner wall of the housing **1211A** and made of plastic material, and the plurality of second ribs **1212B** can be formed around at least one portion of the outer wall of the hub **1212A** and made of plastic material. Furthermore, a cross section of each of the plurality of first rib **1211B** and a cross section of each of the plurality of second rib **1212B** can be formed in arc shapes. In such a way, the first ribs **1211B** and the second ribs **1212B** can be forced to be elastically deflected to disengage from each other for allowing the rotation of the second mount member **1212** relative to the first mount member **1211** when the second tube member **123** is operated by a user. Besides, the first ribs **1211B** and the second ribs **1212B** can be elastically recovered to engage with each other for positioning the second mount member **1212** relative to the first mount member **1211** when the second tube member **123** is released by the user. Therefore, no additional releasing mechanism for disengaging the first ribs from the second ribs is required, which simplifies operation and brings convenience in use.

(13) Please further refer to FIG. **1** to FIG. **5**. In order for easy storage and transportation of the stroller **1**, the frame **11** is foldable and includes a handle **111**, a front foot **112**, a rear foot **113** and a seat **114** installed on the handle **111**. The first mount member **1211** of the tube mount **121** is disposed on the front foot **112**, and the tube mount **121** further includes a third mount member **1213** disposed on the rear foot **113** and rotatably combined with the first mount member **1211** and the handle **113**. In other words, the front foot **112** is rotatably connected to the rear foot **113** by the two tube mounts **121**, and the handle **111** is rotatably connected to the rear foot **113** by the two tube mounts **121**.

(14) In addition, in order to facilitate folding operation of the frame **11**, the frame **11** further includes two link members **115**. Each link member **115** is movably connected to the handle **111** and the first mount member **1211** of the corresponding tube mount **121**. The handle **111** drives the front foot **112** to rotate toward the rear foot **113** by the two link members **115** when the handle **111** rotates relative to the third mount member **1213** to move toward the rear foot **113**. However, the present invention is not limited to this embodiment. For example, in another embodiment, there can be no link member or only one link member located at one lateral side. Alternatively, there can be no third mount member, and the handle can be directly pivoted to the front foot or the rear foot accordingly.

(15) In contrast to the prior art, the present invention utilizes the plurality of first ribs on the first mount member and the plurality of second ribs on the second mount member which are preferably made of the plastic material, to allow or restrain rotating movement of the second mount member relative to the first mount member. Therefore, the second tube member can be positioned at any position between the unfolded position and the folded position by positioning the second mount member relative to the first mount member with engagement of the plurality of first ribs and the plurality of the second ribs, which achieves a purpose of a multistage adjustment of the volume of the storage space. Furthermore, the plurality of first ribs and the plurality of second ribs can be forced to be elastically deflected to disengage from each other for allowing the rotating movement of the second mount member relative to the first mount member when the second tube member is operated to rotate relative to the frame.

(16) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A stroller comprising: a frame, the frame comprising a handle and a link member movably connected to the handle; and a basket disposed on the frame, the basket comprising: a tube mount disposed on the frame; a first tube member connected to the tube mount; a second tube member connected to the tube mount and rotatably switchable relative to the frame between a folded position, an unfolded position, and a plurality of stages between the folded and unfolded positions; and a basket covering disposed on the first tube member and the second tube member, a storage space being enclosed by the basket covering, and a volume of the storage space being adjustable among the plurality of stages by rotation of the second tube member relative to the frame; wherein the tube mount comprises a first mount member connected to the first tube member and movably connected to the link member, and the handle drives the first mount member via the link member, so as to drive the first tube member to rotate by the first mount member when the handle rotates.
2. The stroller of claim 1, wherein the tube mount further comprises a second mount member, the first mount member is disposed on the frame, the second mount member is disposed on the second tube member and rotatably combined with the first mount member, and the second tube member is rotatable relative to the frame by rotation of the second mount member relative to the first mount member.
3. The stroller of claim 2, wherein the first mount member comprises a housing, the second mount member comprises a hub rotatably combined with the housing, a plurality of first ribs is formed on at least one portion of an inner wall of the housing, a plurality of second ribs is formed on at least one portion of an outer wall of the hub, and the first mount member and the second mount member engage with each other by the plurality of first ribs and the plurality of second ribs.
4. The stroller of claim 3, wherein at least one of the plurality of first ribs and the plurality of second ribs is made of plastic material.
5. The stroller of claim 3, wherein a cross section of each of the plurality of first ribs and a cross section of each of the plurality of second ribs are formed in arc shapes.
6. The stroller of claim 1, wherein the frame further comprises a front foot and a rear foot, the front foot is rotatably connected to the rear foot by the tube mount, and the handle is rotatably connected to the rear foot by the tube mount.
7. The stroller of claim 6, wherein the tube mount further comprises a second mount member and a third mount member, the first mount member is disposed on the front foot, the second mount member is disposed on the second tube member and rotatably combined with the first mount member, and the third mount member is disposed on the rear foot and rotatably combined with the

first mount member and the handle.

8. The stroller of claim 7, wherein the handle drives the front foot to rotate toward the rear foot by the link member when the handle rotates relative to the third mount member to move toward the rear foot.

9. A basket adapted for a frame of a stroller, the frame comprising a handle and a link member movably connected to the handle, the basket comprising: a tube mount disposed on the frame; a first tube member connected to the tube mount; a second tube member connected to the tube mount and rotatably switchable relative to the frame between a folded position, an unfolded position, and a plurality of stages between the folded and unfolded positions; and a basket covering disposed on the first tube member and the second tube member, a storage space being enclosed by the basket covering, and a volume of the storage space being adjustable among the plurality of stages by rotation of the second tube member relative to the frame; wherein the tube mount comprises a first mount member connected to the first tube member and movably connected to the link member, and the handle drives the first mount member via the link member, so as to drive the first tube member to rotate by the first mount member when the handle rotates.

10. The basket of claim 9, wherein the tube mount further comprises a second mount member, the first mount member is disposed on the frame, the second mount member is disposed on the second tube member and rotatably combined with the first mount member, and the second tube member is rotatable relative to the frame by rotation of the second mount member relative to the first mount member.

11. The basket of claim 10, wherein the first mount member comprises a housing, the second mount member comprises a hub rotatably combined with the housing, a plurality of first ribs is formed on at least one portion of an inner wall of the housing, a plurality of second ribs is formed on at least one portion of an outer wall of the hub, and the first mount member and the second mount member engage with each other by the plurality of first ribs and the plurality of second ribs.

12. The basket of claim 11, wherein at least one of the plurality of first ribs and the plurality of second ribs is made of plastic material.

13. The basket of claim 11, wherein a cross section of each of the plurality of first ribs and a cross section of each of the plurality of second ribs are formed in arc shapes.

14. The basket of claim 9, wherein the tube mount further comprises a second mount member and a third mount member, the first mount member is disposed on a front foot of the frame, the second mount member is disposed on the second tube member and rotatably combined with the first mount member, and the third mount member is disposed on a rear foot of the frame and rotatably combined with the first mount member and the handle of the frame.

15. A stroller comprising: a frame comprising a handle, a front foot, a rear foot, and a link member; and a basket disposed on the frame, the basket comprising: a tube mount disposed on the frame, a first tube member connected to the tube mount, a first mount member disposed on the front foot and connected to the first tube member, a second tube member connected to the tube mount and rotatably switchable relative to the frame between a folded position and an unfolded position, a second mount member disposed on the second tube member and rotatably combined with the first mount member, and a basket covering disposed on the first tube member and the second tube member, a storage space being enclosed by the basket covering, wherein the front foot is rotatably connected to the rear foot by the tube mount, and wherein the handle is rotatably connected to the rear foot by the tube mount, and wherein the link member is movably connected to the handle and the first mount member, and the handle drives the front foot to rotate toward the rear foot by the link member when the handle rotates.

16. The stroller of claim 15, wherein the tube mount further comprises a third mount member disposed on the rear foot and rotatably combined with the first mount member and the handle.

17. The stroller of claim 16, wherein the handle drives the front foot to rotate toward the rear foot by the link member when the handle rotates relative to the third mount member to move toward the

rear foot.

18. The stroller of claim 15, wherein the volume of the storage space of the basket is adjustable by rotation of the second tube member relative to the frame.
